

**REPORT
ON
SOIL INVESTIGATION FOR CONSTRUCTION
OF
WIND MILL
AT
DN 142, BUKKANHATTI VILLAGE, KARNATAKA**

Client

**M/s. Serentica Renewables India Pvt. Ltd,
Karnataka.**

**By
Time**

**TIME INSTITUTE FOR MATERIALS TESTING
NO. 25, SANJEEVI NAGAR,
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MAY 2026

SUMMARY

Project	:	300MW Koppal Wind Project at DN 142, Bukkanhatti Village, Karnataka
Project Reference	:	Time / Soil Test /May 2026
Client	:	M/s. Serentica Renewables India Pvt. Ltd, Karnataka.
Type of Structure	:	Wind Mill Tower
Location	:	DN 142, Bukkanhatti Village, Karnataka
Scope	:	Geotechnical investigation for the entire area of construction
Test conducted	:	One explanatory bore hole up to 10m depth Conducting standard penetration tests, collecting soil samples and laboratory classification test etc.

FINDINGS

Soil at shallow depth	:	Gravels exists up to 4.5m depth
Hard stratum	:	Hard rock exists between 4.5m to 10m depth
Ground water table	:	Ground water was not encountered during the investigation up to 10m depth.

FOUNDATION	:	Based on the soil condition, it is recommended to provide Raft foundation at a depth of 2.475m.
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SOIL INVESTIGATION



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Time Institute for Materials Testing
टाइम इन्स्टिट्यूट चीजे शोध के लिए
Soil & Structural Consultant, Construction Materials Testing Lab



May 07, 2026

To

M/s. Serentica Renewables India Pvt. Ltd,
Karnataka.

REPORT ON SOIL INVESTIGATION

(Ref: TIME/ SOIL TEST/ MAY 2026)

Sub: Soil Investigation for Construction of a wind mill tower at DN 142, Bukkanhatti Village,
Karnataka - reg.

Ref: Based on mail, Dated: 10.12.2024

1.1 Introduction:

It is proposed to construct a **Wind Mill Tower at DN 142, Bukkanhatti Village, Karnataka.** A detailed soil investigation was suggested for estimating the safe bearing capacity of the soil, so that suitable type of foundation can be decided for the proposed structure. Time Institute for Materials Testing (TIME) has been requested to undertake the soil investigation at the proposed site. The study was carried out by TIME between **01.05.2026 & 02.05.2026**. This report presents complete investigation data and discusses the results to recommend the bearing capacity values at different depths for the proposed structure.

Section 2 describes the soil profile and bore log details of one exploratory bore hole. Section 3 of this report presents the grain size distribution curves. Section 4 of this report presents the safe bearing capacity calculation. Section 5 of this report presents the safe Bearing capacity of soil for Raft foundation (60mm Settlement). Section 6 of this report presents the RQD (Rock Quality Designation). Section 7 of this report presents the soil & rock sample photos. Section 8 of this report presents site photos.

07:05:2026

Preamble on Soil Investigation

- 1. Place of Investigation** : Bukkanhatti Village, Karnataka
- 2. Location** : DN 142
- 3. Date of Investigation** : 01.05.2026 & 02.05.2026
- 4. Carried out in the person of** : Mr. Y. Hemanth Kumar
- 5. Site Address** : Bukkanhatti Village, Karnataka
- 6. Client** : M/s. Serentica Renewables India Pvt. Ltd,
Karnataka.
- 7. Reference** : Based on mail dated: 10.12.2024
- 8. Carried out by** : Time Institute for Materials Testing,
Soil and Structural Division,
No. 25, Sanjeevi Nagar,
Trichy - Chennai Bypass Road,
Tiruchirappalli, Tamil Nadu - 620 002.
- 9. Purpose of Soil Investigation** : Construction of wind mill

Soil samples were collected at different depths and were preserved and transported to the laboratory for test.

1.3 Investigation Programme, Procedure and Data:

Investigation was carried out in one location at the site through exploratory bore hole. Soil samples were collected from bore hole at different depths for conducting laboratory test. Standard Penetration test was conducted at regular intervals and 'N' values were recorded. Undisturbed soil samples were collected and were preserved and transported to the laboratory for detailed identification tests. Based on the field and laboratory tests on the samples collected, the results are furnished in this report.

An exploratory bore hole was advanced from the existing ground level using truck mounted rotary drilling techniques supplemented by Bentonite mud circulation. This drilling procedure with mud circulation is found most suitable for making exploratory bore hole. The mud circulation was employed through the drill rods and letting it out through the side jets provided in the cutting tool thus preventing any disturbance at the bore hole bottom. Mud circulation was used to stabilize the sides and the bottom of the bore hole, and then to bring the soil cuts to the surface. It is important to note that the mud jet is not used to cut the soils as in the case of wash boring technique. Use of drilling mud will also help in preventing the disturbance to the soil at the bore hole bottom during drilling operations. Diameter of the bore hole is about 150mm.

Bore hole was always kept full with the drilling mud so that a positive head is maintained in the bore hole thus preventing any disturbance to the soil within the test zone.

The field tests included Standard penetration test, Disturbed and undisturbed soil sampling, identification of different soil layers, Ground water table observation, complete logging of the bore hole, etc.

Laboratory investigation consisted of classification tests like grain size distribution analysis, determination of liquid and plastic limit values etc. Unconfined compression and Direct shear test was conducted on preserved soil samples. All the field and laboratory tests were conducted according to the procedures stipulated in relevant IS codes.

1.4 Review of Field and Laboratory Results:

Sub soil Profile

The original ground level in the presently investigated area was plain terrain.

The natural soil below the existing ground level is gravel. This layer is extended up to 4.5m depth. Below which hard rock exists from 4.5m to 10m depth from the existing ground level. The soil stratification and its variation under present investigation are available in bore log details. The 'N' value is progressively increasing towards depth, due to the soil stratification.

Ground Water Table

Ground water was not encountered during the investigation up to 10m depth.

Shear Strength

Standard Penetration test blow counts 'N' is measured at different levels in the bore hole. Mainly these 'N' values are used to assess the shear strength of different soil layers.

Cohesive layers: The classification of 'N' values in moderate to high plasticity medium clay deposits developed by Terzaghi is used for estimating the unconfined compression strength and undrained shear strength of clay and sandy clay layers. The consistency between the N values and the liquidity index values is reviewed while assigning undrained shear strength for soft to medium clay layers.

Non - Cohesive Layers: The gravel layers towards depth are of residual type and suitable modifications are made to take care of the presence of plastic fines in the residual soil. Conventional method of estimating the angle of shearing resistance ϕ corresponding to the relative density and the grain size distribution as per the classification suggested by Peck and suitably modified by the R.K. Hough to account for the influence of grain size distribution is adopted here.

Depositional Strata of Sub soil based on SPT Values:

Table – (i) Co-relation of SPT value for Cohesionless Soil

SPT (N) Value	Consistency of Soil
0 - 4	Very Loose
4 - 10	Loose
10 - 30	Medium
30 - 50	Dense
> 50	Very Dense

Table – (ii) Co-relation of SPT value for Cohesive Soil

SPT (N) Value	Consistency of Soil
0 - 2	Very Soft
2 - 4	Soft
4 - 8	Medium
8 - 15	Stiff
15 - 30	Very Stiff
> 30	Hard

1.5 Conclusions & Recommendations:

Shear strength parameters and compressibility parameters of different soil layers are established from the field and laboratory tests.

Based on the existing soil condition and its SBC values, it is recommended to provide Raft foundation at a depth of 2.475m from the existing ground level.

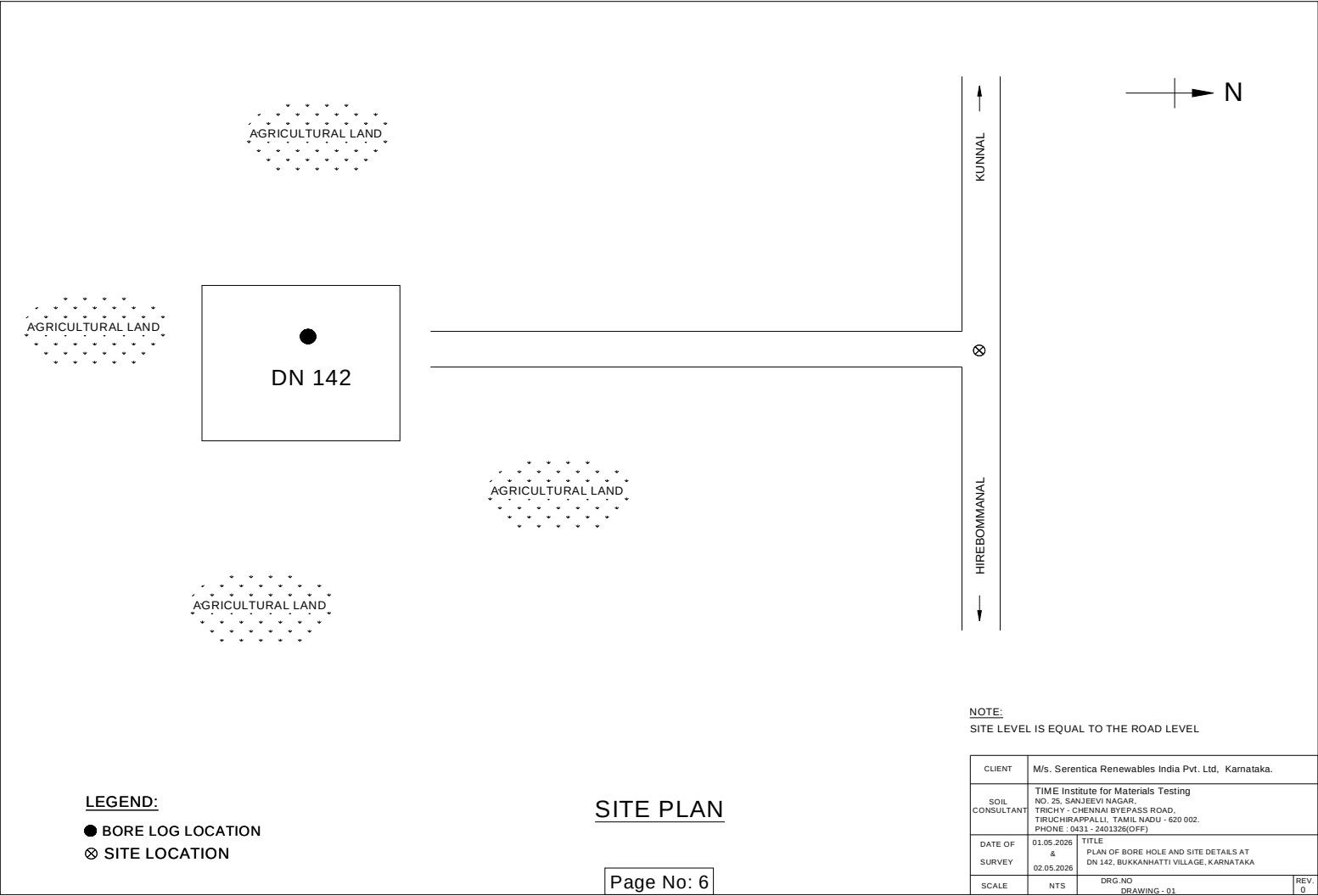
For Time Institute for Materials Testing



Dr. B.N. Krishnaswami.

07th May 2026

SITE LOCATION PLAN



BORE LOG DETAILS



2.3 BORE LOG DETAILS FOR BORE HOLE 1

Project Name		: Report on soil investigation for construction of Wind Mill Tower for					
		M/s. Serentica Renewables India Pvt. Ltd, Karnataka					
Location		: DN 142, Bukkanhatti Village, Karnataka					
Co- ordinates		: 43P E 634616 N 1723874			Type of Boring : Calyx Drill		
Depth of G.W.T		: Nil			Dia of Boring : 150mm		
Site Level		: Equal to the road level			Hammer Wt. & Drop : 63.5 kg/760 mm		
Date of Commencem		: 01.05.2026			Date of Completion : 02.05.2026		
Seismic Zone		: Zone II (AS per IS 1893.1.2002 - Fifth revision)					
Description						Lithology of Bore Hole	Graphical representation of penetration Resistance
Depth below G.L. (m)	Sample Type & Number	Sample Blows per 30cm		Soil Profile	Classification		
1.0	SPT - 1	50	4.5		GP	Gravels	
2.0	SPT - 2	52					
2.475	SPT - 3	53					
3.0	SPT - 4	54					
4.0	SPT - 5	56					
4.5							
10.0	REC = 60%, RQD = 60%		10.0		HR	Hard rock	

Bore terminated at 15m depth from the existing ground level

Legend:

SPT : Standard Penetration Test

GP : Poorly graded gravels

2.3 Summary of Engineering Geo Properties for Bore Hole 1

Density:

The mass density or density of a material is its mass per unit volume. The symbol most often used for density is ρ . Mathematically, density is defined as mass divided by volume:

$$\rho = \frac{m}{V},$$

where

ρ is the density (kg/m^3),

m is the mass (kg), and

V is the volume (m^3)

Relative Density:

Relative density or density index is the ratio of the difference between the void ratios of cohesion less soil in its loosest state and existing natural state to the difference between its void ratio in the loosest and densest states.

$$D_r = \frac{v_{d\max}}{v_d} \times \frac{v_d - v_{d\min}}{v_{d\max} - v_{d\min}} \times 100 \%$$

Poisson's ratio:

Poisson's ratio is the ratio of the transverse strain normal to the applied load, to the axial strain in the direction of the applied load.

Poisson's Ratio can be expressed as

$$\mu = - \epsilon_t / \epsilon_l$$

where

μ = Poisson's ratio; ϵ_t = transverse strain; ϵ_l = longitudinal or axial strain

Strain can be expressed as

$$\epsilon = dl/L$$

where

dl = Change in length; L = Initial length

Young's modulus:

Young's modulus, also known as the **tensile modulus** or **elastic modulus**, is a measure of the stiffness of an elastic material and is a quantity used to characterize materials. It is defined as the ratio of the stress along an axis over the strain along that axis in the range of stress.

Within the limits of elasticity, the ratio of the linear stress to the linear strain is termed the modulus of elasticity or Young's Modulus. It is this property that determines how much a bar will sag under its own weight or under a loading when used as a beam within its limit of proportionality.

The SI unit of modulus of elasticity (E , or less commonly Y) is the pascal (Pa or N/m^2 or $\text{m}^{-1} \cdot \text{kg} \cdot \text{s}^{-2}$). The practical units used are megapascals (MPa or N/mm^2) or gigapascals (GPa or kN/mm^2).

Young's modulus, E , can be calculated by dividing the tensile stress by the tensile strain in the elastic (initial, linear) portion of the stress-strain curve:

$$E = \frac{\text{tensile stress}}{\text{tensile strain}} = \frac{\sigma}{\varepsilon} = \frac{F/A_0}{\Delta L/L_0} = \frac{FL_0}{A_0\Delta L}$$

where

E is the Young's modulus (modulus of elasticity)

F is the force exerted on an object under tension;

A_0 is the original cross-sectional area through which the force is applied;

ΔL is the amount by which the length of the object changes;

L_0 is the original length of the object.

Modulus of Rigidity (or Shear Modulus):

Modulus of Rigidity (or Shear Modulus) is the coefficient of elasticity for a shearing force. It is defined as "the ratio of shear stress to the displacement per unit sample length (shear strain)".

shear modulus or **modulus of rigidity**, denoted by G , is defined as the ratio of shear stress to the shear strain

$$G \stackrel{\text{def}}{=} \frac{\tau_{xy}}{\gamma_{xy}} = \frac{F/A}{\Delta x/l} = \frac{Fl}{A\Delta x}$$

where

$\tau_{xy} = F/A$ = shear stress;

F is the force which acts

A is the area on which the force acts

in engineering, $\gamma_{xy} = \Delta x/l = \tan \theta$ = shear strain. Elsewhere, $\gamma_{xy} = \theta$

Δx is the transverse displacement

l is the initial length

Shear modulus' derived SI unit is the pascal (Pa), although it is usually expressed in gigapascals (GPa).

Bulk Modulus:

The **bulk modulus** (K or B) of a substance measures the substance's resistance to uniform compression. It is defined as the ratio of the infinitesimal pressure increase to the resulting *relative* decrease of the volume. Its base unit is the pascal.

Bulk modulus $K > 0$ can be formally defined by the equation

$$K = -V \frac{dP}{dV}$$

where P is pressure, V is volume, and dP/dV denotes the derivative of pressure with respect to volume. Equivalently

$$K = \rho \frac{\partial P}{\partial \rho}$$

where ρ is density and $dP/d\rho$ denotes the derivative of pressure with respect to density.

Compressive Strength:

Maximum stress a material can sustain under crush loading. The compressive strength of a material that fails by shattering fracture can be defined within fairly narrow limits as an independent property. However, the compressive strength of materials that do not shatter in compression must be defined as the amount of stress required to distort the material an arbitrary amount. Compressive strength is calculated by dividing the maximum load by the original cross-sectional area of a specimen in a compression test.

Compressive strength = Load / Area

$$\sigma = \frac{F}{A}$$

where, F = Load applied [KN], A = Area [m²]

Drained Shear Strength:

The drained shear strength is the shear strength of the soil when pore fluid pressures, generated during the course of shearing the soil, are able to dissipate during shearing. It also applies where no pore water exists in the soil (the soil is dry) and hence pore fluid pressures are negligible. It is commonly approximated using the Mohr-Coulomb equation. (It was called "Coulomb's equation" by Karl von Terzaghi in 1942.) (Terzaghi 1942) combined it with the principle of effective stress.

In terms of effective stresses, the shear strength is often approximated by:

$$\tau = \sigma' \tan(\phi') + c'$$

Where $\sigma' = (\sigma - u)$, is defined as the effective stress. σ is the total stress applied normal to the shear plane, and u is the pore water pressure acting on the same plane.

ϕ' = the effective stress friction angle, or the 'angle of internal friction' after Coulomb friction. The coefficient of friction μ is equal to $\tan(\phi')$. Different values of friction angle can be defined, including the peak friction angle, ϕ'_p , the critical state friction angle, ϕ'_{cv} , or residual friction angle, ϕ'_r .

c' = is called cohesion, however, it usually arises as a consequence of forcing a straight line to fit through measured values of (τ, σ') even though the data actually falls on a curve. The intercept of the straight line on the shear stress axis is called the cohesion. It is well known that the resulting intercept depends on the range of stresses considered: it is not a fundamental soil property. The curvature (nonlinearity) of the failure envelope occurs because the dilatancy of closely packed soil particles depends on confining pressure.

Undrained Shear Strength:

This term describes a type of shear strength in soil mechanics as distinct from drained strength.

Conceptually, there is no such thing as *the* undrained strength of a soil. It depends on a number of factors, the main ones being:

- Orientation of stresses
- Stress path
- Rate of shearing
- Volume of material (like for fissured clays or rock mass)

Undrained strength is typically defined by Tresca theory, based on Mohr's circle as:

$$\sigma_1 - \sigma_3 = 2 S_u$$

Where:

σ_1 is the major principal stress

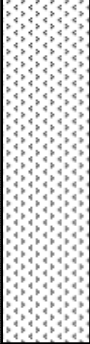
σ_3 is the minor principal stress

S_u is the shear strength $(\sigma_1 - \sigma_3)/2$

hence, $\tau = S_u$ (or sometimes c_u), the undrained strength.

It is commonly adopted in limit equilibrium analyses where the rate of loading is very much greater than the rate at which pore water pressures, that are generated due to the action of shearing the soil, may dissipate. An example of this is rapid loading of sands during an earthquake, or the failure of a clay slope during heavy rain, and applies to most failures that occur during construction.

As an implication of undrained condition, no elastic volumetric strains occur, and thus Poisson's ratio is assumed to remain 0.5 throughout shearing. The Tresca soil model also assumes no plastic volumetric strains occur. This is of significance in more advanced analyses such as in finite element analysis. In these advanced analysis methods, soil models other than Tresca may be used to model the undrained condition including Mohr-Coulomb and critical state soil models such as the modified Cam-clay model, provided Poisson's ratio is maintained at 0.5.

Lithology of Bore Hole			Natural Moisture Content (%)	Bulk Density (gm/cc)	Density (kg/m ³)	Elasticity Modulus (Mpa)	Modulus Rigidity (MPa)	Bulk Modulus (MPa)	Poisson ratio
Depth below (m)	G.L.	Profile Soil Description							
0 - 4.5m		 Gravels	9.45	1.99	1818	790	319	506	0.24
4.5m - 10m		 Hard rock	0.001	2.40	2400	2041	850	1134	0.20

2.4. DYNAMIC PROPERTIES OF SOIL

Elastic Modulus (E static):

The modulus of elasticity (also known as the elastic modulus, the tensile modulus or Young's modulus) is a number that measures an object or substance resistance to being deformed elastically (i.e., non-permanently) when a force is applied to it. The elastic modulus of an object is defined as the slope of its stress – strain curve in the elastic deformation region. A stiffer material will have a higher elastic modulus.

Static young's modulus E_s , which can be obtained from the following Equation

$$E_{\text{static}} (E_s) = \text{Linear Stress} / \text{Linear Strain}$$

Where stress is the force causing the deformation divided by the area to which the force is applied and strain is the ratio of the change in length parameter caused by the deformation to the original value of the length parameter.

Elastic Modulus (E dynamic):

E_{dynamic} which can be obtained from the following formula

$$E_{\text{dynamic}} (E_d) = 2 (1 + \mu) G'$$

Where

μ = Poisson's Ratio for dynamic condition

G' = Dynamic Shear Modulus

Poisson's Ratio (μ):

Poisson's Ratio (μ) is defined as the ratio of axial compression (ϵ_v) to lateral expansion (ϵ_L) strains.

$$\begin{aligned}\mu &= 1/m \\ m &= \epsilon_v / \epsilon_L\end{aligned}$$

Therefore
$$\mu = \epsilon_L / \epsilon_v$$

Shear Modulus (G'):

Shear Modulus (G') is defined as the ratio of Shear Stress to Shear Strain. It is related to E_d and μ .

$$\text{i.e. } G' = \frac{E_d}{2(1 + \mu)}$$

Stiffness:

Stiffness is the rigidity of an object – the extent to which it resists deformation in response to an applied force. The complementary concept is flexibility or pliability: the more flexible an object is the less stiff it is.

The Stiffness, k , of a body is a measure of the resistance offered by an elastic body to deformation. For an elastic body with a single degree of freedom (DOF) (for example, stretching or compression of a rod), the stiffness is defined as

$$k = F / \delta$$

where,

F = The force applied on the body

δ = The displacement produced by the force along the same degree of freedom (for instance, the change in length of a stretched spring)

In the International system of Units, Stiffness is typically measured in newtons per meter.

Rotational Stiffness:

A body may also have a rotational stiffness, k is given by

$$k = M / \theta$$

Where

M = Applied moment

θ = Rotation.

In the SI system, rotational stiffness is typically measured in Newton – meters per radian
The formula to calculate rotational stiffness for circular foundation will be

$$k_r = \frac{8 G' R^3}{3 (1 - \mu)}$$

Where

k_r = Rotational Stiffness

G' = Dynamic Shear Modulus

R = Radius of foundation

μ = Poisson's Ratio

Dynamic Spring Constant, k:

Springs have their own natural "spring constants" that define how stiff they are. The letter k is used for the spring constant. By Newton's Third Law of Motion, as a spring is pulled, it pulls back with a restoring force.

Vertical mode:

$$k_z = \frac{4G r_o \eta_z}{1 - \mu}$$

Where

$$\begin{aligned} k_z &= \text{Dynamic Spring Constant} \\ G' &= \text{Dynamic Shear Modulus} \\ r_o &= \text{Radius of foundation} \\ \mu &= \text{Poisson's Ratio} \\ \eta_z &= 1 + 0.6 (1 - \mu) (h/r_o) \end{aligned}$$

Where

$$h = \text{Depth of embedment}$$

Horizontal mode:

$$\begin{aligned} k_z &= \frac{32 (1 - \mu) G r_o \eta_x}{7 - 8\mu} \\ \eta_x &= 1 + 0.55 (2 - \mu) (h/r_o) \end{aligned}$$

Rocking mode:

$$\begin{aligned} k_\phi &= \frac{8 G r_o^3 \eta_\phi}{3(1 - \mu)} \\ \eta_\phi &= 1 + 1.2 (1 - \mu) (h/r_o) + 0.2 (2 - \mu) (h/r_o)^3 \end{aligned}$$

Torsion mode:

$$k_\psi = \frac{16 G r_o^3}{3}$$

Summary of Dynamic Properties for soil

Lithology of Bore Hole			Youngs Modulus of Elasticity, E _{Dynamic} (Mpa)	Poisson's Ratio (μ) for dynamic condition	Shear Modulus (G') (Mpa)	Dynamic Spring Constant (N/m)				Rotational Stiffness (Nm/rad)
Depth below G.L. (m)	Profile	Soil Description				Vertical Mode (N/m)	Horizontal Mode (N/m)	Rocking Mode (Nm)	Torsion Mode (Nm)	
0 - 4.5m		Gravels	3138	0.34	1171	7.793.E+10	7.084.E+10	5.683.E+12	6.245.E+06	4.73E+12
4.5m - 10m		Hard rock	6232	0.30	2397	1.512.E+11	1.437.E+11	1.108.E+13	1.278.E+07	9.13E+12

2.5 Summary of Laboratory Test Results for Bore Hole 1

S.No.	Bore hole No.	Depth (m)	Soil Description	Relative Density (%)	Specific gravity	Gravel (%)	Coarse sand (%)	Medium Sand (%)	Fine sand (%)	Silt (%)	Clay (%)
1	Bore Hole 1	1.0	Gravels	55	2.66	53.6	26.7	16.5	3.2	0.0	0.0
2		2.0		57	2.66	54.2	27.5	15.4	2.9	0.0	0.0
3		2.475		59	2.66	55.9	28.1	13.7	2.3	0.0	0.0
4		3.0		60	2.66	57.6	29.5	10.8	2.1	0.0	0.0
5		4.0		61	2.66	59.5	30.2	8.7	1.6	0.0	0.0

2.6 Shear Strength Parameters for Different layers of Bore Hole 1

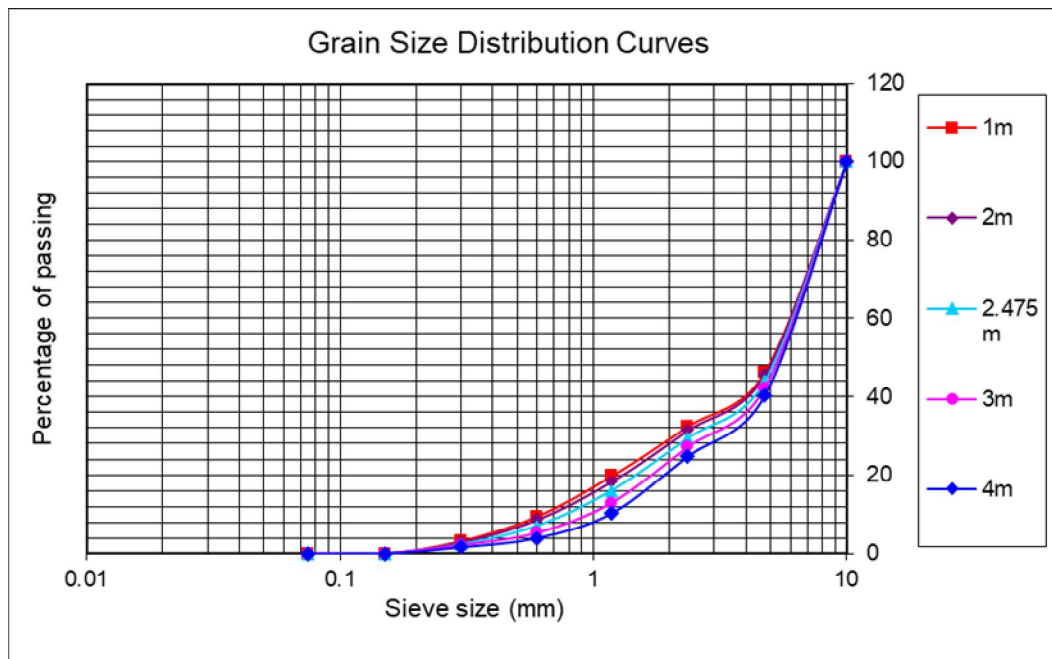
S.No.	Bore Hole No.	Depth below G.L. (m)	Soil Description	N	Shear Strength ' ϕ ' (in degrees)
1	Bore Hole 1	1.0	Gravels	50	33.93
2		2.0		52	34.16
3		2.475		53	34.38
5		3.0		54	34.61
6		4.0		56	34.83

GRAIN SIZE ANALYSIS

3.1 TEST REPORT ON SIEVE ANALYSIS OF SAMPLE

Bore Hole – 1 @ 1m to 4m

SIEVE SIZE	PERCENTAGE OF PASSING				
	1m Depth	2m Depth	2.475m Depth	3m Depth	4m Depth
10 mm	100	100	100	100	100
4.75 mm	46.4	45.8	44.1	42.4	40.5
2.36 mm	32.5	31.5	29.5	27.2	24.8
1.18 mm	19.7	18.3	16	12.9	10.3
600 micron	9.5	8.7	7.1	5.3	4.0
300 micron	3.2	2.9	2.3	2.1	1.6
150 micron	0	0	0	0	0
75 micron	0	0	0	0	0



Description	1m Depth	2m Depth	2.475m Depth	3m Depth	4m Depth
Fineness modulus	4.887	4.928	5.010	5.101	5.188

Observation:

The inference from the graph shows that the gravels exists up to 4m depth.

CALCULATION
OF
SAFE BEARING CAPACITY

4.1 Safe Bearing Capacity calculations based on shear consideration (Direct Shear):

As Per IS 6403:1981

At 2.475m depth (Bore Hole 1)

Proving Ring capacity 200kg = 966.2 division

Area of the shear box = 6cm x 6cm = 36cm²

Normal Stress (kg/cm ²)	No. of Division	Stress
0.5	24	0.138
1.0	83	0.477
1.5	143	0.822

$$\begin{aligned}\text{Stress} &= \text{Load} / \text{Area} \\ &= (200/966.2) * 24 * (1/36) = 0.138 \\ \phi &= \tan^{-1}(0.822-0.138) = 34.38^\circ\end{aligned}$$

Net safe bearing capacity of the soil (as per IS 6403:1981)

General Shear failure

$$q_{ult} = C N_c S_c d_c i_c + q (N_q - 1) S_q d_q i_q + 0.5 B \gamma N_\gamma S_\gamma d_\gamma i_\gamma W'$$

(Intermediate Shear failure)

$$q_{ult} = (5/6) C N'_c S_c d_c i_c + q (N'_q - 1) S_q d_q i_q + 0.5 B \gamma N'_\gamma S_\gamma d_\gamma i_\gamma W'$$

(Local Shear failure)

$$q_{ult} = (2/3) C N'_c S_c d_c i_c + q (N'_q - 1) S_q d_q i_q + 0.5 B \gamma N'_\gamma S_\gamma d_\gamma i_\gamma W'$$

Where

C	=	Cohesion in t/m ²
N _c , N _q , N _γ	=	Bearing Capacity Factors for angle of internal friction (Φ)
N' _c , N' _q , N' _γ	=	Bearing Capacity Factors for angle of internal friction (Φ')
Where, Φ'	=	tan ⁻¹ (0.67 tan Φ)
N'' _c , N'' _q , N'' _γ	=	Bearing Capacity Factors for Intermediate values of Φ & Φ'
S _c , S _q , S _γ	=	Shape Factors (From Table 2 of IS 6403 – 1981)
d _c , d _q , d _γ	=	Depth Factors
i _c , i _q , i _γ	=	Inclination Factors
q	=	Effective surcharge at the base level of foundation in t/m ²
B	=	Width of footing, diameter of circular footing in m
γ	=	Bulk unit weight of soil in t/m ³
W'	=	Correction factor for location of water table

At 2.475m depth (Bore hole 1), the Soil profile is Gravels

$$\begin{aligned} \text{If } C &= 0 \\ \Phi &= 34.38^\circ \text{ (based on laboratory test)} \end{aligned}$$

This depth belongs to Intermediate shear failure

Applying these parameters in the above equation

$$\begin{aligned} q_{ult} &= (5/6) C N''_c S_c d_c i_c + q (N''_q - 1) S_q d_q i_q + 0.5 B \gamma N''_v S_v d_v i_v \\ W' \\ \text{Where } C &= 0 \text{ (Since no Cohesion)} \\ \text{Therefore, } q_{ult} &= q (N''_q - 1) S_q d_q i_q + 0.5 B \gamma N''_v S_v d_v i_v W' \end{aligned}$$

$$\begin{aligned} N''_c &= 31.823; \\ N''_q &= 20.489; \\ N''_v &= 26.866 \end{aligned}$$

Bearing Capacity calculation through shear density of soil ' γ '

There are two relationships of weight of soil to Volume of soil as

- i) Total density or Bulk density
- ii) Dry density

$$\text{Therefore, Total Bulk density, } \gamma_t = \frac{\text{Total Weight of soil}}{\text{Total Volume of soil}} = \frac{W}{V}$$

$$\text{Dry Density } \gamma_d = W_s/V; \quad \text{Whereas } W_s = \text{Total weight of solids}$$

In a particular case the total density will be known as saturated density, when the soil is saturated.

Another density of soil will be known as submerged or buoyant density, γ_b when the soil is submerged in water.

$$\gamma_b = \gamma_t - \gamma_w; \quad \gamma_w = \text{Density of water} = 1 \text{ g/cc or } 1 \text{ t/m}^3$$

Mostly soils submerged in water are saturated.

$$\text{Hence } \gamma_{\text{buoyant}} = \gamma_{\text{sat}} - \gamma_w$$

Based on the above equations,

$$\begin{aligned} \gamma &= \gamma_{\text{sat}} = 1.99 \text{ t/m}^3 \\ d &= 2.475\text{m} \\ \text{Here, } \gamma_b &= \gamma_{\text{sat}} - 1 = 1.99 - 1 = 0.99 \text{ t/m}^3 \end{aligned}$$

$$\begin{aligned}\text{Therefore, } q &= \gamma_b \times d = 0.99 \times 2.475 = 2.450 \text{ t/m}^2 \\ S_q &= 1.2 \quad \& S_\gamma = 0.6 \\ d_q &= 1.023 \quad \& d_\gamma = 1.023 \\ i_\gamma &= 1 \quad \& i_q = 1 \\ B &= \text{Width / Diameter of foundation} = 20\text{m} \\ W' &= 0.5\end{aligned}$$

Applying in the above equation

$$\begin{aligned}q_{ult} &= q (N_q - 1) S_q d_q i_q + 0.5 B \gamma N_\gamma S_\gamma d_\gamma i_\gamma W' \\ &= 2.450 \times (20.489 - 1) \times 1.2 \times 1.023 \times 1 + 0.5 \times 20 \times 1.99 \times 26.866 \times 0.6 \times 1.023 \times 1 \times 0.5 \\ &= 222.80 \text{ t/m}^2\end{aligned}$$

Applying factor of safety as 3

Then allowable bearing capacity

$$q_{na} = 222.80 / 3 = 74.27 \text{ t/m}^2$$

4.2 Safe Bearing Capacity calculations based on settlement consideration

(SPT value):

Elastic Settlement Calculation

Considering dia of the raft (d)	=	20 m
Considering thickness of the raft (t)	=	1.5 m
Area of the raft $(\pi \times d^2)/4$	=	314.159 m ²
Considering the circular dia of foundation as equivalent size of square portion (as per Bowel's Page.No.306)		
Therefore side of the square, L (or) B	=	17.725 m
B'	=	8.8623 m
$\frac{1}{4}$ {Total depth of Investigation (10m) - Founding depth (2.475m)}	=	7.525 m
Therefore H/B'	=	0.849
L/B	=	1

Therefore I_1 , & I_2 from table 5.2 of page no. 304 of Bowle's

$$I_1 = 0.1133$$

$$I_2 = 0.0830$$

I_s (From page 306 of

$$\text{Poisson value of the soil Bowle's book)} = I_1 + [(1-2\nu)/(1-\nu)]I_2$$

Soil Strata from Below foundation to Investigation depth				Elasticity Modulus, 'E' (Mpa)	Poisson ratio ' μ '
Soil Strata	From	To	Depth (m)		
Gravels	2.475	4.500	2.025	790	0.24
Hard Rock	4.500	10.000	5.500	2041	0.2
Average				1704.352	0.21

$$\text{Poisson value of the soil } \nu = 0.21$$

$$I_s = 0.174$$

$$m \text{ (Since the load is at centre from page 306)} = 4$$

$$I_f \text{ (Based on IS 8009 part 1-1976)} = 0.95$$

$$\text{The self weight of the foundation} = \pi/4 \times d^2 \times t \times \rho$$

Considering submerged unit weight of concrete,

$$\rho = 1.5 \text{ t/m}^3$$

$$= 706.86 \text{ t}$$

$$= 707.00 \text{ t (nearly)}$$

$$\text{Approximate total vertical load including dynamic component on} = 150 \text{ t}$$

$$\text{Hence the total vertical load on soil } Q = 857 \text{ t}$$

Therefore Soil pressure at the base of the footing as per Bowel's page No. 288

$$q_v = (Q/Z^2) A_b$$

$$\text{where } Z = \text{Founding Depth (m)} = 2.475 \text{ m}$$

$$\text{When } r \text{ (i.e., Load at the centre of the foundation)} = 0$$

$$r/z \text{ from the table in the page 288 of bowel's} = 0$$

$$A_b \text{ is a function of } r/z = 0.477 \text{ (for } r/z = 0)$$

$$q_v = 66.73 \text{ t/m}^2$$

Hence the elastic immediate settlement ΔH (as per Bowel's page no. 306)

$$\text{Where, } E_{\text{average}} = 170435.22 \text{ t/m}^2$$

$$\Delta H = q_v B' [(1-\mu^2) / E_{\text{average}}] \times m \times I_s \times I_f$$

$$= 0.00219657 \text{ m}$$

$$= 2.20 \text{ mm}$$

Calculation of safe bearing capacity for Raft or Mat foundation:

For 60mm Settlement:

$$q = 0.7 (N - 3) R'_w C_d S_p$$

Where,

$$q = \text{Net safe bearing capacity of soil (t/m}^2\text{)}$$

$$S_p = \text{Permissible settlement (6.0 cm)}$$

$$N = \text{Corrected SPT values}$$

$$R'_w = \text{Correction factor for water table level}$$

$$C_d = \text{Correction factor for depth of embedment}$$

$$= 1 + (D/B)$$

$$D = \text{Depth} = 2.475\text{m}$$

$$B = \text{Width / Diameter of foundation} = 20\text{m}$$

$$C_d = 1 + (2.475/20) = 1.1237$$

Then $q_{na} = 0.7 (N - 3) R'_w C_d \times 6.0$

This location (DN 142) @ 2.475m depth SPT 'N' Value = 53

$$\Rightarrow q_{na} = 0.7 \times (53 - 3) \times 0.5 \times 1.1237 \times 6.0 = 117.99 \text{ t/m}^2$$

CALCULATION
OF
SAFE BEARING CAPACITY VALUES
FOR
RAFT FOUNDATION
(60MM SETTLEMENT)

5. Calculation of safe bearing capacity of the soil for Raft foundation:

5.1 Safe Bearing capacity of soil based on SPT values

Bore Hole No.	Depth (m)	SPT value	Safe Bearing capacity of soil (60mm Settlement)	
			KN/m ²	t/m ²
BH – 1	1.00	50	1036.4	103.64
BH – 1	2.00	52	1131.9	113.19
BH – 1	2.475	53	1179.9	117.99
BH – 1	3.00	54	1231.4	123.14
BH – 1	4.00	56	1335.6	133.56

5.2 Safe Bearing capacity of soil based on Shear Failure Criteria:

Through Direct Shear

Bore Hole No.	Depth (m)	Normal Stress (kg/cm ²)	Division	Ø (in degrees)	Safe Bearing Capacity of Soil (t/m ²)
BH -1	1.00	0.5 1.0 1.5	24 82 141	33.93°	57.55
BH -1	2.00	0.5 1.0 1.5	24 82 142	34.16°	67.62
BH -1	2.475	0.5 1.0 1.5	24 83 143	34.38°	74.27
BH -1	3.00	0.5 1.0 1.5	24 84 144	34.61°	81.38
BH -1	4.00	0.5 1.0 1.5	24 84 145	34.83°	93.19

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5.3 Calculation of Bearing Capacity through shear considering both Angle of internal friction & Cohesion

Bore Hole No.	Depth (m)	Soil Profile	Input Values								Mode of Failure	Bearing capacity factor			Shape Factor			Depth Factor			Inclination Factor			Net Ultimate Bearing Capacity (q_u t/m ²)	Safe Bearing Capacity (q_s t/m ²)
			Dia (m)	W'	γ_{soil}	C	ϕ	ϕ'	F	q (t/m ²)		N_c	N_q	N_γ	S_c	S_q	S_γ	d_c	d_q	d_γ	$S_{c'}$	$S_{q'}$	$S_{\gamma'}$		
DN 142	1.0	Gravels	20.00	0.50	1.98	0.00	33.93	24.16	3	0.98	Intermediate	30.718	19.475	25.136	1.3	1.2	0.8	1.019	1.009	1.009	1	1	1	172.64	57.55
	2.0		20.00	0.50	1.98	0.00	34.16	24.34	3	1.96	Intermediate	31.276	19.986	26.004	1.3	1.2	0.8	1.038	1.019	1.019	1	1	1	202.87	67.62
	2.475		20.00	0.50	1.99	0.00	34.38	24.52	3	2.45	Intermediate	31.823	20.489	26.866	1.3	1.2	0.8	1.047	1.023	1.023	1	1	1	222.80	74.27
	3.0		20.00	0.50	1.99	0.00	34.61	24.71	3	2.97	Intermediate	32.410	21.032	27.801	1.3	1.2	0.8	1.057	1.029	1.029	1	1	1	244.15	81.38
	4.0		20.00	0.50	1.99	0.00	34.83	24.88	3	3.96	Intermediate	32.985	21.567	28.729	1.3	1.2	0.8	1.077	1.038	1.038	1	1	1	279.56	93.19

Legend

L = Length; B = Breadth; W' = Correction factor for water table; γ_{soil} = Unit weight of soil; C = Cohesion;

ϕ = Angle of shearing resistance (General Shear); ϕ' = Angle of shearing resistance (Local Shear);

F = Factor of Safety; q = Effective surcharge of soil above footing;

5.4 Comparison of safe bearing capacity of soil through Shear and Settlement consideration:

Bore Hole no.	Depth (m)	Safe Bearing Capacity (t/m ²)	
		Shear consideration (Direct Shear)	Settlement consideration (60mm Settlement)
BH – 1	1.00	57.55	103.64
BH – 1	2.00	67.62	113.19
BH – 1	2.475	74.27	117.99
BH – 1	3.00	81.38	123.14
BH – 1	4.00	93.19	133.56

5.5 Recommended safe bearing capacity values at different depths for

Raft Foundation

Considering the lesser value of the above, the safe Bearing Capacity, values at different depths are recommended below:

Bore Hole	Depth (m)	Safe Bearing Capacity (t/m²) (60mm Settlement)
BH – 1	1.00	57.55
BH – 1	2.00	67.62
BH – 1	2.475	74.27
BH – 1	3.00	81.38
BH – 1	4.00	93.19

The above test data and findings are relevant and applicable only for the soil conditions existing at the test site. Any change in the soil strata observed during excavation may be reported to the laboratory, which will redefine the bearing capacity of the soil.

RQD
(ROCK QUALITY DESIGNATION)

6. RQD (ROCK QUALITY DESIGNATION)

6.1 Definition of RQD

The RQD is defined as the cumulative length of core pieces longer than 10cm in a run divided by the total length of the core run. The total length of core must include all lost core sections. Any mechanical breaks cause by the drilling process or in extracting the core from the core barrel should be ignored. RQD should be measured as soon as possible after drilling.

Table 1. Rock Mass Classification Based on RQD

RQD	Rock Quality Classification
<25%	Very Poor
25-50%	Poor
50-75%	Fair
75-90%	Good
90-100%	Excellent

6.2 Geotechnical Parameters for Rock Sample

Water Absorption:

This test helps to determine the water absorption of rock sample as per IS 2720 – Part – II. Two sample pieces should be tested separately and the individual and mean results should be reported.

$$\text{Water Absorption} = (A - B) / B * 100$$

Where,

A is the Weight of the Saturated surface dried sample

B is the Weight of the Oven dried sample

Point Load Index Test and Uniaxial Compressive Strength Test of Rock

The International Society of Rock Mechanics (ISRM, 1985) has established the basic procedures for testing and calculation of the point load strength index. There are three basic types of Point load tests: axial, diametral and block or lump. The axial and diametral tests are conducted on rock core samples. In the axial test, the core is loaded parallel to the longitudinal axis of the core and this test is most comparable to a UCS test.

Point Load Index test consists of squeezing pieces of rock diametrically between two hardened steel cones. For these tests, rock pieces must have a length at least 1.4 times their diameter. The Point load test allows the determination of the uncorrected Point load strength index. It must be corrected to the equivalent diameter (De) of 54mm. If the core being tested is “near” 54mm in diameter (like NX core), the correction is not necessary.

The point load index, I_s is equal to

$$I_s = P / D^2$$

Where

P is the load at rupture and

D is the core diameter

As Hoek (1977) pointed out, the mechanics of the PLT actually causes the rock to fail in tension. The PLT's accuracy in predicting the UCS therefore depends on the ratio between the UCS and the tensile strength. For most brittle rocks, the ratio is approximately 10. For soft mudstones and clay stones, however, the ratio may be closer to 5. This implies that PLT results might have to be interpreted differently for the weakest rocks.

Early studies (Bieniawski, 1975; Broach and Franklin, 1972) were conducted on hard, strong rocks and found that relationship between UCS and the Point load strength could be expressed as

$$UCS = (K) I_{s50} = 24 I_{s50}$$

Where

K is the "conversion factor". Subsequent studies found that $K = 24$ was not as universal as had been hoped and that instead there appeared to be a broad range of conversion factors.

6.3 Rock Mass Rating System for Jointed Rock Masses (Bieniawski)

A. Classification Parameters and their Ratings

Parameter			Range of Values						
1.	Strength of intact rock material	Point – load Strength Index	> 10 MPa	4-10 MPa	2-4 MPa	1-2 MPa	For this low range uniaxial compressive test is preferred		
		Uniaxial Compressive Strength	> 250 MPa	100-250 MPa	50-100 MPa	25-50 MPa	5-25 MPa	1-5 MPa	< 1 MPa
	Rating		15	12	7	4	2	1	0
2.	Drill core quality RQD		90%-100%	75%-90%	50%-75%	25%-50%	< 25%		
	Rating		20	17	13	8	3		
3.	Spacing of discontinuities		> 2m	0.6 – 2m	200-600mm	60-200mm	< 60mm		
	Rating		20	15	10	8	5		
4.	Condition of discontinuities		Very rough surfaces. Not continuous No separation	Slightly rough surfaces. Separation < 1mm. Slightly Weathered walls.	Slightly rough surfaces. Separation < 1mm. Highly Weathered walls.	Slickenside surfaces or Gouge < 5mm or Separation 1-5mm continuous	Soft gouge > 5mm thick or Separation > 5mm continuous		
	Rating		30	25	20	10	0		
5.	Ground water	Inflow per 10m tunnel length	None (OR)	<10 litres/min (OR)	10-25 litres/min (OR)	25-125 litres/min (OR)	>125 litres/min (OR)		
		Joint water Ratio pressure may or principal stress	0 (OR)	0,0 – 0,1 (OR)	0,1 – 0,2 (OR)	0,2 – 0,5 (OR)	>0,5 (OR)		
		General conditions	Completely dry	Damp	Wet	Dripping	Flowing		
	Rating		15	10	7	4	0		

6.4 GEOTECHNICAL PARAMETERS FOR ROCK SAMPLE

AT 4.5m to 10m DEPTH

The following laboratory test results are obtained for the rock sample Report on soil investigation for construction of Wind Mill Tower at DN 142, Bukkanhatti Village, Karnataka site at a depth of 4.5m to 10m (Hard rock). The rock core photos are presented in the soil and rock sample photos.

S.No.	IS Codes	Parameter	Result (4.5m to 10m depth)
1	IS 11315 Part 11 - 1985	REC	60 %
2	IS 11315 Part 11 - 1985	RQD	60 %
3	IS 9143 - 1979	Uniaxial Compressive strength of rock	42 Mpa
4	IS 8764 - 1998	Point Load Index	1.75 Mpa
5	IS 2720 Part II	Water Absorption	Nil
6	(IS 1124 – 1974)	Density	3.400 gm/cc
7	(IS 1124 – 1974)	Specific gravity	2.85
8	(IS 11315 Part 11 – 1985)	RMR	72%
9	(IS 9143 – 1979)	Allowable Bearing Pressure	372 t/m ²
10	Description of rock		Good

SOIL & ROCK SAMPLE PHOTOS



Bore Hole No: 1

Depth : 0 to 10m

SITE PHOTOS



DN 142, Bukkanhatti Village, Karnataka